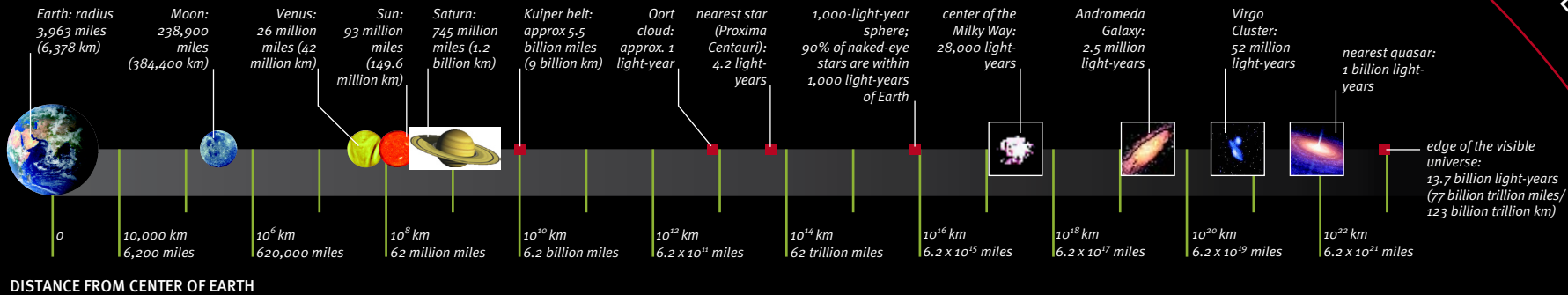


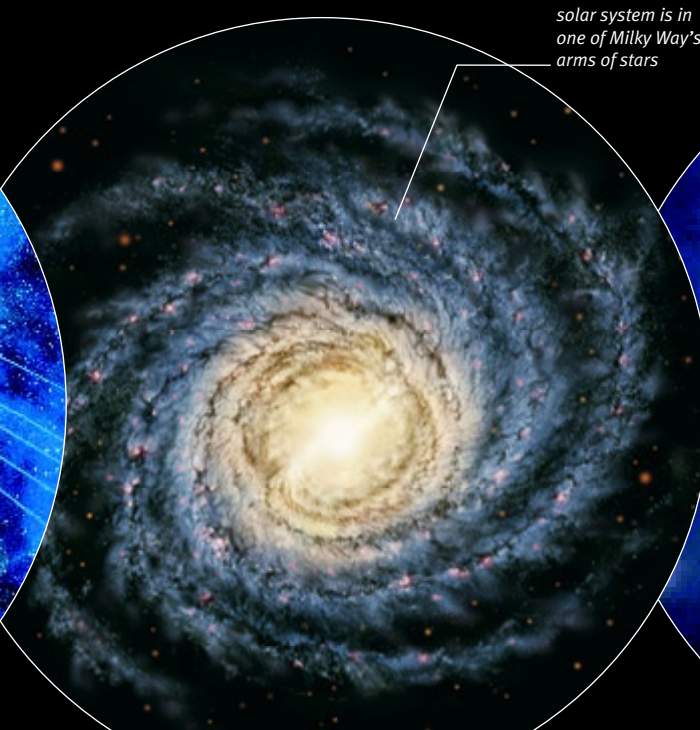
MEASURING DISTANCE

The universe is so vast that the units of measurement used on Earth, such as miles, quickly become inadequate. They are used to describe the size of the planets within the solar system, as well as distances between them, but beyond the solar system, the unit commonly used is the light-year (ly). One light-year is 5.88 trillion miles (9.46 trillion km) and is the distance that light travels in a year. Light moves at 186,282 miles per second (299,792 km per second), and

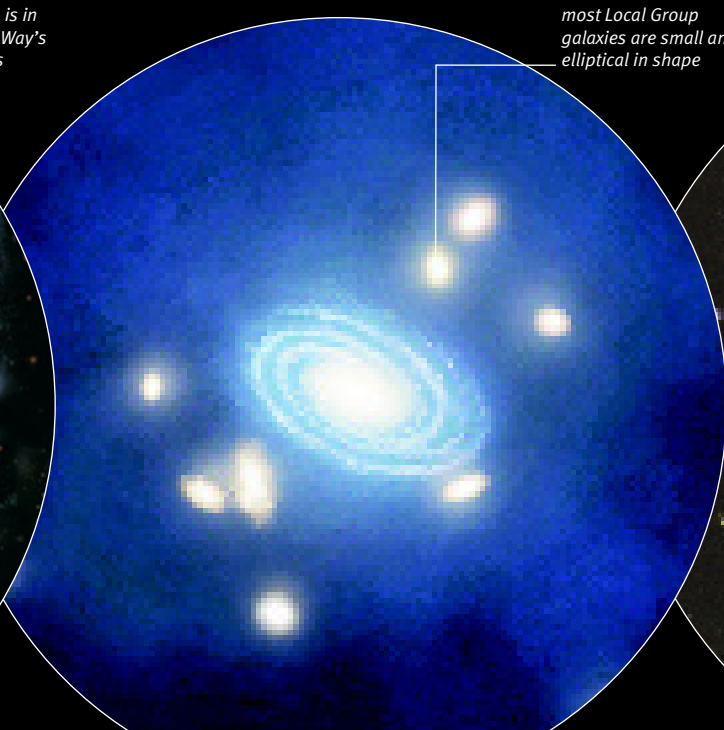
nothing in the universe travels faster. The Andromeda Galaxy is said to be 2.5 million light-years away, because it takes that length of time for its light to reach us; this means we are seeing the galaxy as it was 2.5 million years ago. By contrast, the Sun's light takes just eight-and-a-half minutes to reach Earth. In the chart below, the first division represents 10,000 km (6,200 miles). Each further division marks a 10x increase in scale on the previous one.



LOOKING INTO SPACE



solar system is in one of Milky Way's arms of stars



most Local Group galaxies are small and elliptical in shape

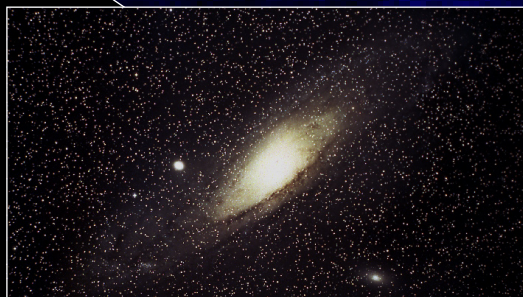


about 10,000 galaxies appear in this image by the Hubble Space Telescope



The Milky Way Galaxy

The Sun is just one of billions of stars in the Milky Way Galaxy (above top). The stars are in a disk shape, with a concentration of stars in the center, and "arms" of stars spiraling out from it. All the stars we see in the night sky are in the Milky Way, along with clusters of stars (above) and giant clouds of gas and dust where new stars form.



The Local Group of galaxies

The Milky Way is one of a group of more than 40 galaxies that exist together in space. They are collectively called the Local Group (above top). The Milky Way and the Andromeda Galaxy (above) dominate the group. The Andromeda Galaxy is about 2.5 times as wide as the Milky Way and is the most distant object normally visible to the naked eye.



Universe of galaxies

In whatever direction we look and however deep we peer into space, there are galaxies (above top)—an estimated 100–125 billion of them. They exist in clusters (above) that are strung together into superclusters, which are the biggest structures in the universe.